INCH-POUND

MIL-C-83723/66A(USAF)
17 December 1991

SUPERSEDING
MIL-C-83723/66(USAF)
8 May 1975

MILITARY SPECIFICATION SHEET

(A) CONNECTORS, ELECTRICAL, (CIRCULAR, ENVIRONMENT RESISTING), PLUG, THREADED COUPLING, (PUSH-PULL, QUICK-DISCONNECT WITHOUT LANYARD), CRIMP PIN CONTACT, SERIES III, CLASS A, G, R, AND W)

(A) This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

(A) The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specific in the solicitation: MIL-C-83723.

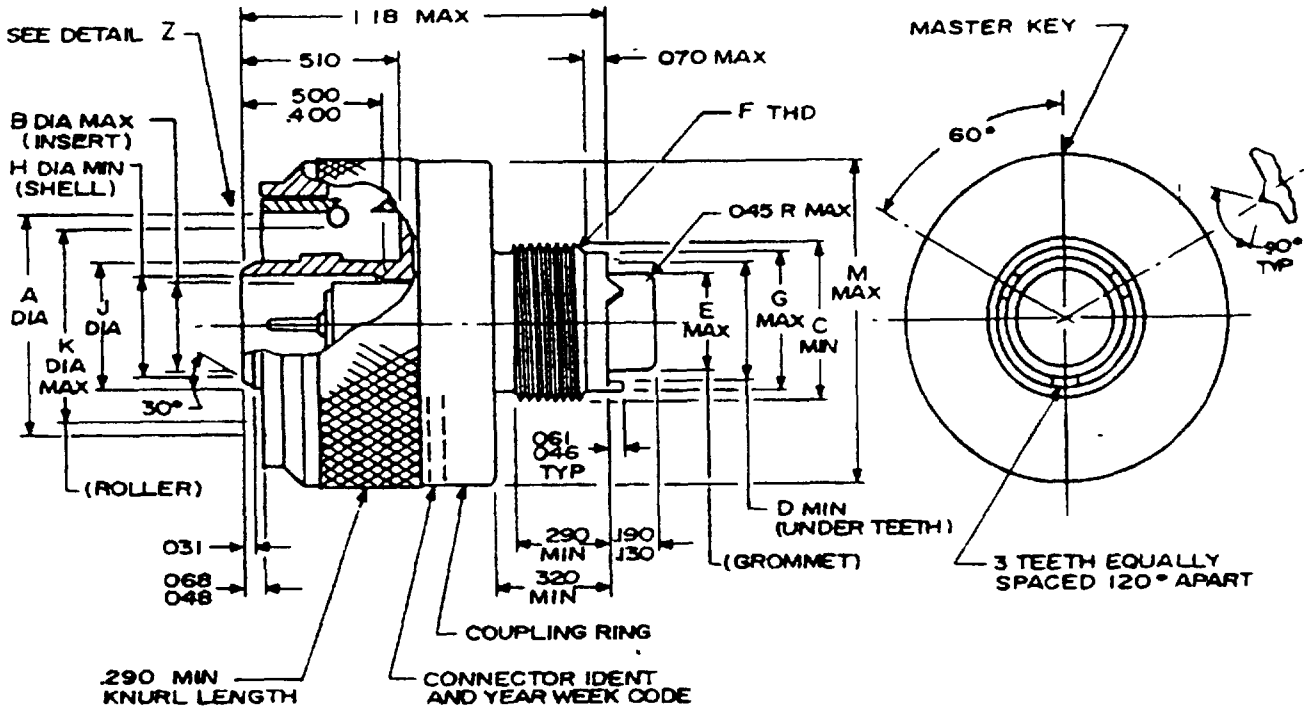
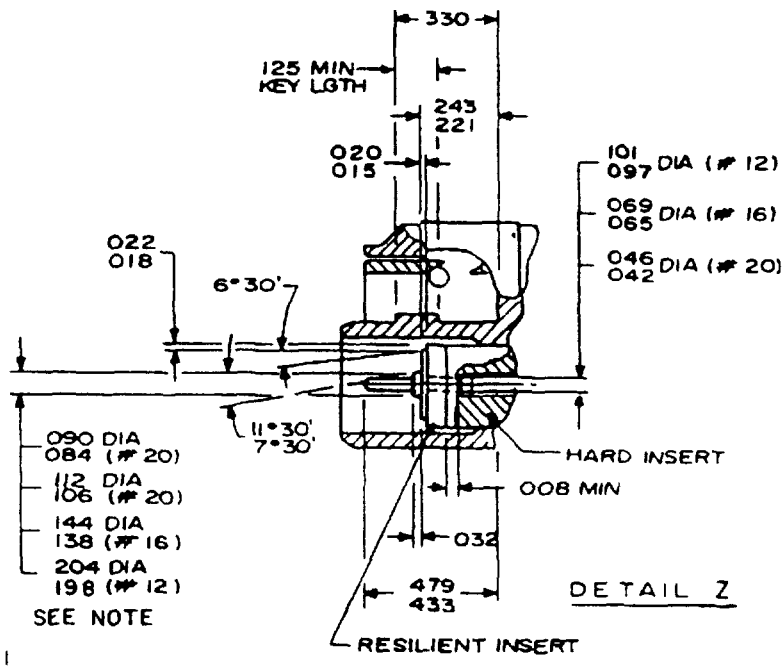


FIGURE 1. Dimensions and configurations.

(A) denotes changes

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Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
008	0.20	.045	1.14	.084	2.13	.144	3.66	.330	8.38
.015	0.38	.046	1.17	.090	2.29	.190	4.83	.400	10.16
.018	0.46	.048	1.22	.101	2.57	.198	5.03	.433	11.00
.020	0.51	.061	1.55	.106	2.69	204	5.18	.479	12.17
.022	0.56	.065	1.65	112	2.84	.221	5.61	500	12.70
.031	0.79	.068	1.73	.125	3.18	.243	6.17	510	12.95
.032	0.81	.069	1.75	.130	3.30	.290	7.37	1.18	29.90
.042	1.07	.070	1.78	.138	3.51	.320	8.13		

FIGURE 1. Dimensions and configurations - Continued.

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Connector dimensions.

Shell size	A	B	C	D	E	F thread	G	H
8	.704	.290	.470	.370	.305	.500-20 UNF-2A	.437	.294
10	.829	.388	.600	.497	.405	.625-24 UNEF-2A	.572	.428
12	1.016	.558	.724	.613	.531	.750-20 UNEF-2A	.687	.598
14	1.079	.627	.849	.738	.665	.875-20 UNEF-2A	.812	.667
16	1.204	.772	.974	.863	.790	1.000-20 UNEF-2A	.937	.794
18	1.345	.860	1.030	.919	.869	1.0625-18 UNEF-2A	.992	.900
20	1.470	.985	1.154	1.044	.994	1.1875-18 UNEF-2A	1.117	1.025
22	1.595	1.110	1.279	1.169	1.119	1.3125-18 UNEF-2A	1.242	1.150
24	1.720	1.235	1.404	1.294	1.244	1.4375-18 UNEF-2A	1.367	1.275

FIGURE 1. Dimensions and configurations - Continued.

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Connector dimensions - Continued.

Shell size	J	K	M
8	.405 .396	.612	1.095
10	.526 .517	.737	1.240
12	.696 .687	.925	1.432
14	.765 .756	987	1 490
16	.892 .883	1.112	1.711
18	.998 .989	1.237	1 815
20	1.123 1.114	1 362	1.962
22	1.248 1 239	1.487	2.070
24	1.373 1.364	1 612	2 195

FIGURE 1. Dimensions and configurations - Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.015	0.38	.225	5.72	.600	15.24	.898	22.81	1.235	31.37
.018	0.46	.243	6.17	.612	15.54	.900	22.86	1.237	31.42
.020	0.51	.282	7.16	.613	15.57	.919	23.34	1.239	31.47
.022	0.56	.285	7.24	.625	15.88	.925	23.50	1.240	31.50
.031	0.79	.290	7.37	.627	15.93	.937	23.80	1.242	31.55
.032	0.81	.294	7.47	.643	16.33	.974	24.74	1.248	31.70
.042	1.07	.305	7.75	.646	16.41	.985	25.02	1.275	32.39
.045	1.14	.320	8.13	.665	16.89	.987	25.07	1.279	32.49
.046	1.17	.367	9.32	.667	16.94	.992	25.20	1.294	32.87
.048	1.22	.370	9.40	.687	17.45	.994	25.25	1.313	33.34
.054	1.37	.378	9.60	.696	17.68	.998	25.35	1.345	34.16
.061	1.55	.388	9.86	.704	17.88	1.000	25.40	1.362	34.59
.065	1.65	.396	10.06	.705	17.91	1.016	25.81	1.364	34.65
.068	1.73	.400	10.16	.708	17.98	1.025	26.04	1.367	34.72
.069	1.75	.402	10.21	.724	18.39	1.044	26.52	1.373	34.87
.070	1.78	.428	10.87	.730	18.44	1.063	27.00	1.432	36.37
.081	2.06	.433	11.00	.737	18.72	1.079	27.40	1.438	36.51
.084	2.13	.437	11.10	.738	18.75	1.090	27.69	1.470	37.34
.087	2.21	.465	11.81	.750	19.05	1.095	27.81	1.487	37.77
090	2.29	.468	11.89	.756	19.20	1 110	28.19	1.490	37.85
097	2.46	.470	11.94	.765	19.43	1.112	28.24	1.595	40.51
.101	2.57	.479	12.17	.772	19.61	1.114	28.40	1.612	40.94
106	2.69	.481	12.22	.790	20.07	1.117	28.37	1.711	43.46
112	2.84	.497	12.62	.794	20.17	1.119	28.42	1.720	43.69
.118	3.00	.500	12.70	.812	20.62	1.123	28.52	1.815	46.10
.130	3.30	.510	12.95	.829	21.06	1.130	28.70	1.962	49.83
138	3.51	.526	13.36	.849	21.56	1.150	29.21	2.070	52.58
.144	3.66	.531	13.49	.860	21.84	1.154	29.31	2.129	56.62
.190	4.38	.558	14.17	.863	21.92	1.169	29.69	2.195	55.75
.198	5.03	.572	14.53	.869	22.07	1.180	30.16	2.353	59.77
.204	5.18	.580	14.73	.875	22.23	1.188	32.89		
.221	5.61	.583	14.81	.883	22.43	1.204	30.58		
.222	5.64	.598	15.19	.892	22.66	1.224	31.09		

. NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance shall be .XX ±.010 and .XX ±.005, angular, X° ±1°0 ' and X°X' ±30'.
4. The dimensions showing relationship of coupling ring to shell apply only when coupling ring is in full forward position.
5. The quick-disconnect connector plug is intended to be used with series III threaded coupling receptacle (MIL-C-83723/82 and /84) with adapter (MIL-C-83723/70) permanently attached to the receptacle.
6. Coupling of the quick-disconnect plug connector is to be accomplished by a straight axial push with no cocking of the coupling mechanism.
7. Uncoupling of the quick-disconnect plug connector is to be accomplished by a straight axial pull on the outer coupling ring.

FIGURE 1. Dimensions and configurations - Continued.

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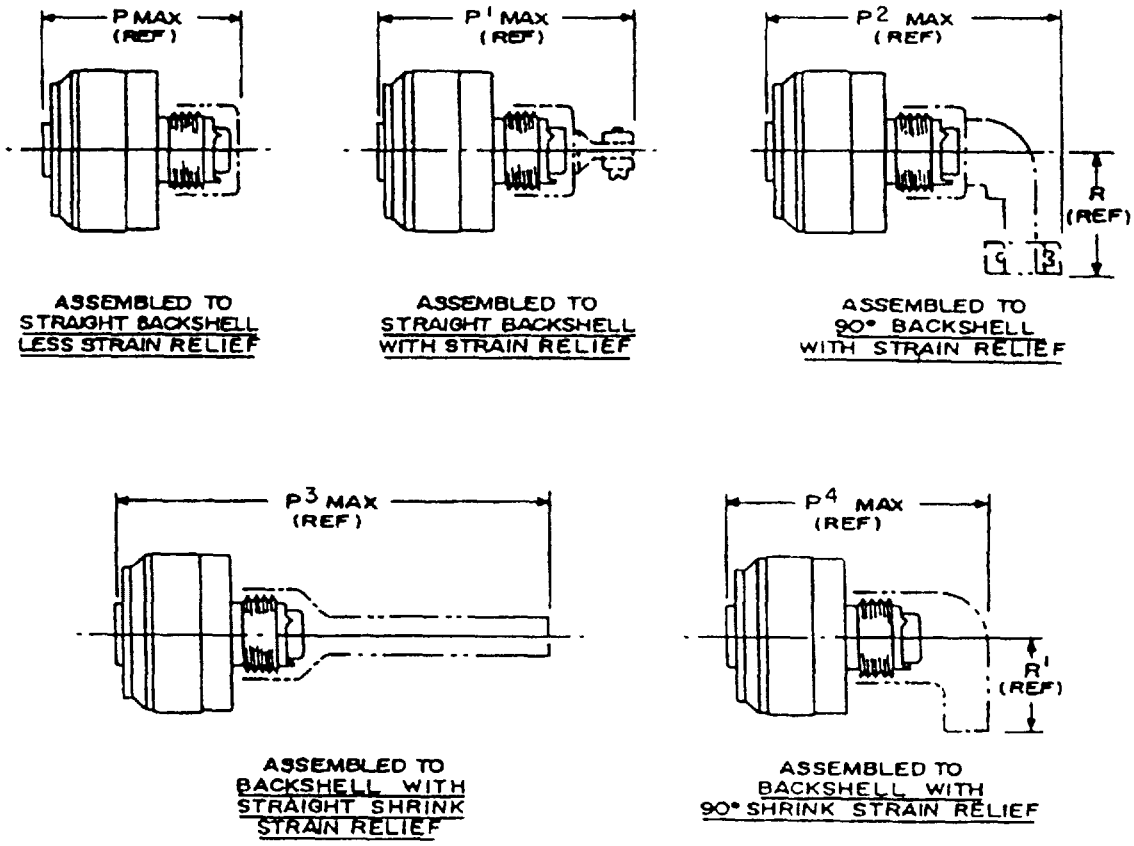


FIGURE 2. Backshell.

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Backshell dimensions.

Shell size	P	P1	P2	P3	P4	R	R1
8	1.457	1.869	2.018	5.838	1.848	.830	4.585
10	1.457	1.979	2.100	6.528	1.888	.880	5.390
12	1.457	2.124	2.293	7.348	1.928	.950	6.330
14	1.457	2.124	2.353	7 348	1 928	1 010	6.330
16	1.457	2.229	2.502	8.068	2.028	1.070	6.710
18	1.457	2.474	2.614	8.068	2 028	1.130	6.710
20	1.457	2.604	2.739	9.068	2.148	1.190	7.740
22	1.457	2.729	2.864	9 068	2.148	1 260	7.740
24	1.457	2.854	2.989	9.068	2.228	1 320	8.030

Inches	mm	Inches	mm	Inches	mm
.830	21.08	2.018	51.26	2.854	72.49
.880	22.35	2.028	51.51	2.864	72.49
.950	24.13	2.100	53.34	2.989	72.75
1.010	25.65	2.124	53 95	4 585	116.46
1.070	27.18	2.148	54.56	5.390	136.91
1.090	27.69	2.228	56.59	5.838	148.29
1.130	28.70	2.229	56 62	6.330	160.78
1.260	32.00	2.293	58.24	6.528	165.81
1.320	33.53	2.353	59.77	6.710	170.43
1.457	37.01	2.474	62.84	7.348	186.64
1.848	46.10	2.502	63.55	7.440	196.60
1.869	47.47	2.604	66.14	8.030	203.96
1.888	47.96	2.614	66.40	8.068	204.93
1.928	48.97	2.729	69.37	9.068	230.32
1.979	50.27	2.739	69.57		

NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance shall be XX ±.010 and XX ±.005, angular, X° ±1°0' and X°X' ±30'.

FIGURE 2. Backshell - Continued.

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TABLE I. Insert arrangements requiring reduced diameters for raised seal barriers ont he outer row of the contact cavities.

Shell size	Insert arrangements	Contact cavities
8	(A) 03	1, 2, and 3

REQUIREMENTS:

Dimensions and configuration: See figure 1 through 3.

For insert arrangement: See MIL-STD-1554 and table I

For connector accessories: See MIL-C-85049 and MIL-C-83723/16(USAF).

Unless otherwise specified in the purchase order.

a. Contacts shall be furnished with the connector. The total quantity shall include 10 percent or 1, whichever is greater, spare contacts of each size contacts used See MIL-C-39029.

b. Grommet seal plugs shall be furnished with the connector. Total quantity shall be 15 percent of each contact size or 3, whichever is greater. See MS27488.

c. The insert/extraction tool for each size contact shall be included in the connector package. See MIL-I-81969.

Connector assembly must include a backshell to meet the moisture sealing requirements. See MIL-C-85049 and MIL-C-83723/16(USAF).

The coupling mechanism of the connector plug shall withstand a minimum tensile load of 100 pounds Load to be applied to rear accessory hardware in a straight axial direction

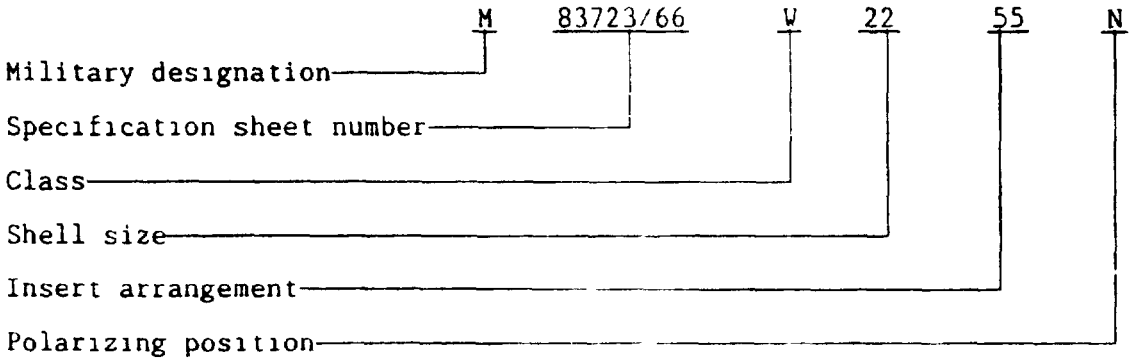
The the quick-disconnect connector plug shall meet the vibration and shock requirements in accordance with paragraph of MIL-C-83723 and the coupling mechanism shall maintain the plug and receptacle in a locked condition.

The tensile force required to unlock the coupling mechanism shall be 5 pounds minimum and 20 lbs maximum

Locking mechanism as required to meet performance requirements of MIL-C-83723.

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(A) Part or Identifying Number (PIN) example:



CONCLUDING MATERIAL

Custodians:
Air Force - 85

Review activities.
Air Force - 11, 70, 80, 84
DLA - ES

Preparing activity
Air Force - 85

Agent:
DLA - ES

(Project 5935-F360-02)

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